

# Digital Electronics

## Assignment-2

- 1) Design a circuit to test whether a 4-bit binary number is a valid BCD code or not.
- 2) Suppose an 7485 IC has its pin for A=B output broken. How can this chip be used for the detection of the equality of the two 4-bit input numbers?
- 3) Design a full-adder circuit using (a) XOR, (b) AND, (c) OR, (d) NAND, (e) NOR, (f) half-adder circuits.
- 4) Design a full-subtractor circuit using (a) XOR, (b) AND, (c) OR, (d) NAND, (e) NOR, (f) half-subtractor circuits.
- 5) Five soldiers A, B, C, D, E volunteer to perform an important military task if their following conditions are satisfied.
  - i. Either A or B or both must go.
  - ii. Either C or E or both must go.
  - iii. Either both A and C go or neither goes.
  - iv. If D goes, then E must also go.
  - v. If B goes, then A and C must also go.
- 6) Convert 1110 (Gray) to binary and give an all XOR realization.
- 7) Perform the arithmetic operation:
  - i)  $(-42) \text{ (decimal)} + (-13) \text{ (decimal)}$  in 2's complement binary form.
  - ii)  $(-42) \text{ (decimal)} + (-13) \text{ (decimal)}$  in 1's complement binary form.
  - iii)  $(-22) \text{ (decimal)} + (13) \text{ (decimal)}$  in 2's complement binary form.
- 8) i) Minimize the following expression in POS form using K-map and realize the simplified function using NAND gates only:  $F(A,B,C,D) = \sum (1,2,3,8,9,10,11,14) + d\sum (7,15)$   
ii) Minimize the following expression using K-map and realize the simplified function using NOR gates only:  $F(A,B,C,D) = M\sum (1,2,3,8,9,10,11,14) + d\sum (7,15)$
- 9) Draw the logic circuit diagram for a 2 to 1 line multiplexer and design a 4 to 1 line multiplexer using the 2 to 1 line multiplexer.
- 10) Design a MOD – 10 synchronous binary Up-counter using D F-F and other gates.
- 11) Simplify the Boolean function
  - i)  $F(A,B,C,D) = \sum (1,4,5,6,7,11,12,13,15)$  using Q-M method in SOP form
  - ii)  $F(A,B,C,D) = \sum (0,1,2,8,10,11,14,15) + \sum d(3,13)$  using Q-M in POS form
- 12) Design a full-adder circuit using minimum number of NAND gates.
- 13) How the half adders can be connected together to make a full adder?
- 14) Design a full sub tractor and adder using only 4:1 MUX.
- 15) What is the difference between a Mealy and Moore type sequential circuit?
- 16) Determine bases following:
  - i.  $\sqrt{41}=5$
  - ii.  $302/20=12.1$
  - iii.  $23+44+14+32=233$
  - iv.  $1234+5432=6666$
- 17) Draw a logic circuit add two BCD numbers to get the BCD sum using binary adders and other necessary logic gates. Explain the operation of the circuit.
- 18) Design a combinational circuit using all discrete logic gates to convert-
  - i) BCD to Excess-3code.
  - ii) Excess-3 to BCD code